

CADEN DAVIS

cdavis9@stanford.edu | 949-606-6648 | San Clemente, CA | www.linkedin.com/in/caden-davis

EDUCATION

Stanford University

Palo Alto, CA

Master of Science in Electrical Engineering (MSEE)

September 2025 – Graduating June 2027

Specialization in Signal Processing (DSP) and Embedded Systems | Breadth Areas in VLSI and ML

University of California, Los Angeles (UCLA)

Los Angeles, CA

Bachelor of Science in Electrical Engineering (BSEE)

September 2021 – June 2025

Cumulative GPA: 4.00/4.00 | ECE Department Valedictorian 2025 | Fast Track to Success ECE Honors Program

EXPERIENCE

Cellular RF Firmware Intern

San Diego, CA

Apple Inc.

June 2025 – September 2025

- Developed internal Python tool for converting erroneous cellular field scenario logs into cycle-accurate C++ transceiver tests
- Auto-generated test cases for cellular integration issues enable reproducibility on internal platforms to verify fixes
- Constructed rule-based database for parsing modem AT commands to track RF hardware allocation and timing dependencies
- Productionized via integration with a developer CLI and Jenkins CI for early detection of timing regressions
- Designed and delivered a tech talk on multi-stage AGC architectures in cellular receivers for cross-functional training

Digital Signal Processing Intern

Irvine, CA

Anduril Industries, Inc.

June 2024 – September 2024

- Optimized detectors for frequency-hopping spread spectrum OFDM signals for 3x greater probability of signal intercept
- Wrote embedded application software in C for radio module control and chirp spread spectrum (CSS) waveform design
- Developed end-to-end physical RF system test platform to verify and quantify efficacy of jamming waveform designs
- Acted as program coordinator to align teams in meeting customer performance targets against high-priority adversaries

Wireless Research Assistant

Los Angeles, CA

Wireless Lab, UCLA ECE

Sept 2023 – June 2025

- Developed mmWave phased array and SDR control platform in Python as lab infrastructure for beamforming experiments
- Implemented range-Doppler radar and IEEE 802.11a (OFDM Wi-Fi) core PHY for joint communication and sensing (JCAS)
- JCAS project for next-generation smart traffic lights awarded \$1.5M in design competition for US Dept of Transportation
- Simulated MAC and PHY layers in MATLAB for networked underwater sensing nodes utilizing acoustic communication
- Developed PCB and C++ interface for real-world sensor node prototype verified in Marina Del Rey harbor
- “LocoMote: AI-driven Sensor Tags for Fine-Grained Undersea Localization and Sensing,” 2nd Author, *IEEE Sensors Journal*

Electrical Hardware Design Intern

Irvine, CA

Anduril Industries, Inc.

June 2023 – September 2023

- Designed schematic and PCB in Altium Designer for vehicle control user interface with DFT / DFM constraints
- Developed bare-metal C firmware on STM32F4 microcontroller to process low-rate user inputs with polling superloop
- Improved robustness with length-matched differential SPI over LVDS, temperature-aware layout, and mil-spec harnessing
- Drove integration efforts with mechanical and firmware teams on enclosure design, shock testing, and vehicle codebase

Embedded Imaging Software Intern

Irvine, CA

Viseon Inc.

June 2021 – September 2021

- Implemented live video enhancement application for minimizing reflective tissue glare using Python, C++, and OpenCV
- Real-time histogram equalization and glare reduction features approved by board of marketing and board of surgeons
- Optimized video application on the Jetson Nano SBC, meeting latency target of sub-120ms for 30fps live camera feed
- Created thread-level process flowcharts to effectively communicate low-level software processes to executive board

PROJECTS

IoT Smart Sleepmask using Multicore-Enabled RTOS | C, FreeRTOS

September 2024 – June 2025

- Built a dual-core ESP32 pipeline with cross-core queues separating sensing and networking in producer-consumer pattern
- Implemented DMA-driven ADC and I2C sensor capture with real-time FIR heart-rate filtering
- Published telemetry over MQTT to AWS IoT Core with buffering, batching, and automatic retry for intermittent links

Out-of-Order Superscalar RISC-V Processor | *Verilog, Intel Quartus Prime*

September 2024 – December 2025

- Implemented an out-of-order, dual-issue core with 2 ALUs and 1 load/store unit with forwarding over a common data bus
- Built Tomasulo-style microarchitectural modules including the reservation station, register alias table, and reorder buffer
- Designed a load-store queue for memory disambiguation and a direct-mapped L1 data cache to reduce memory latency
- Created a functional verification testbench for the RV32I instruction set to validate pipeline and memory correctness

FPGA Audio Spectrum Visualizer | *SystemVerilog, Intel Quartus Prime*

September 2023 – June 2024

- Wrote Cooley-Tukey FFT module with SystemVerilog on Intel FPGA for capturing spectra of microphone input signals
- Developed VGA interface and ping-pong RAM buffering of FFT outputs for smooth 60Hz display of spectrum to a monitor
- Expanded project with signal processing features including Hamming windowing, FIR filtering, and DC offset calibration

From-Scratch Superheterodyne Radio | *C, MATLAB, LTspice*

September 2022 – June 2023

- Implemented BPSK radio DSP pipeline with Costas Loop carrier recovery and Mueller-Muller timing recovery in MATLAB
- Rewrote application in C for real-time processing on STM32F4 microcontroller with STM32CubeIDE
- Designed and simulated amplifiers, mixers, oscillators, and filters for HF band Tx/Rx superheterodyne radio pair in LTspice

COURSEWORK

Computer Architecture (RISC-V) | Operating Systems (POSIX, C) | Digital Logic Design | Data Structures & Algorithms (C++)
Digital Signal Processing I, II | Audio & Image Processing | Computational Imaging | Embedded Systems | Real-World Sensors
Digital Integrated Circuits (Cadence Virtuoso) | VLSI Systems | Analog Integrated Circuits | RF Systems | Electromagnetics I, II
Reinforcement Learning | Wireless Communications I, II | Optimization I, II | Random Processes | Deep Learning